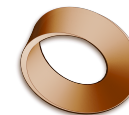


Name: _____
Class: _____
Date: _____



Mr. Rawson • GCSE Physics

Kinematics Graphing — ANSWERS

The Basic 6, with the reasoning underneath. Each page carries one motion type: the three graphs as they should be drawn, and then — under each graph — why it has that shape. Read the explanation boxes across, left to right: they are the same argument told three times, once in each of the three languages a physicist uses to describe motion.

The three rules that answer every one of these

1. On a position–time graph, the slope is the velocity.

Sloping up means moving in the + direction; sloping down means moving in the – direction. A *straight* line means the velocity is not changing. A *curve* means it is. The steeper the line, the faster the object.

2. On a velocity–time graph, the height is the velocity and the slope is the acceleration.

Above the axis, the object is moving in the + direction; below it, in the – direction. A horizontal line means no acceleration at all. A sloping line means the velocity is changing, and how steeply it slopes is how hard.

3. On an acceleration–time graph, the height is the acceleration.

In all six of these motions the acceleration is constant, so every one of these graphs is a horizontal line. That is the whole answer to question 9 on the Physics Classroom sheet.

The one that catches everybody

“Speeding up” does not mean positive acceleration.

It is tempting to read a positive acceleration as “getting faster” and a negative one as “slowing down”. That is wrong, and motions 5 and 6 exist to prove it.

What actually decides it is whether the velocity and the acceleration point the **same way**:

same sign as each other	the object speeds up
opposite signs	the object slows down

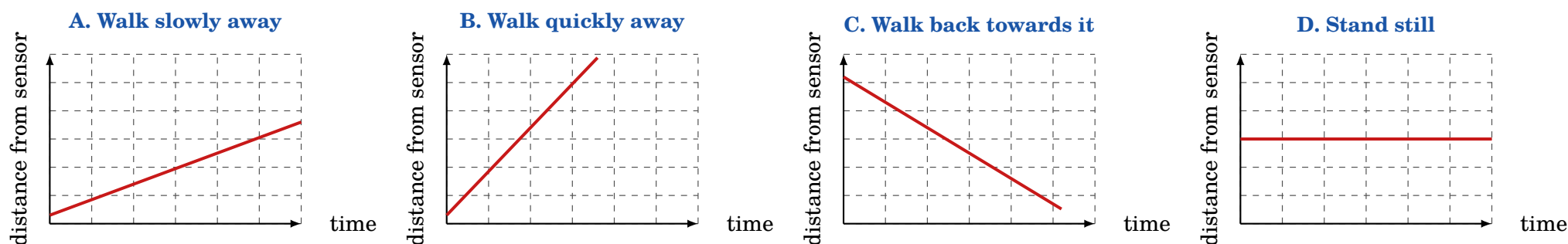
So motion 5 — moving in the – direction and speeding up — has a *negative* acceleration. And motion 6 — moving in the – direction and slowing down — has a *positive* one, even though nothing about it is getting faster.

Read the two graphs together, never one alone.

The numbers below are the values these graphs are drawn for. If the simulation on the day starts from different defaults, the shapes are unchanged — only the numbers on the axes move.
Simulation: [The Physics Classroom — Kinematic Graphing](#).

What somebody has told you about the car	position	velocity	acceleration
1. "It is 30 m past the school gate."	✓		
2. "It is moving at 12 m/s, north."		✓	
3. "It picks up another 2 m/s every second."			✓
4. "It is holding a steady 12 m/s." — which one is zero ?			✓
5. "It is reversing at a steady 3 m/s." — which one is negative ?		✓	
6. "It is slowing from 20 m/s to a stop." — which one is negative ?			✓

The three that do the work are 4, 5 and 6. 4 — a steady speed is *not* a zero velocity; it is a zero *acceleration*, because nothing is changing. 5 — the velocity is negative because of the *direction*; the car is not slowing down, and its acceleration is zero. 6 — here the acceleration is negative and the velocity is still positive. Lines 5 and 6 are the whole lesson in words, before a single graph is drawn.



what the four walks show

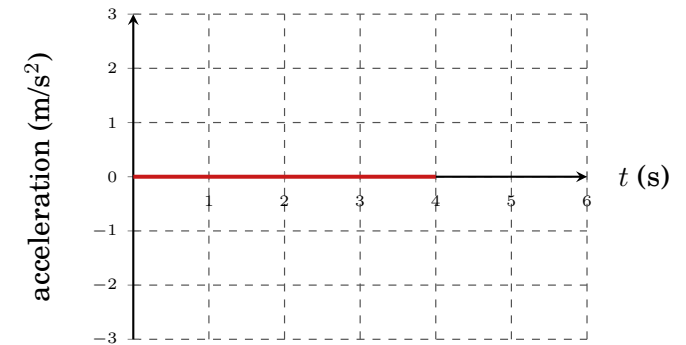
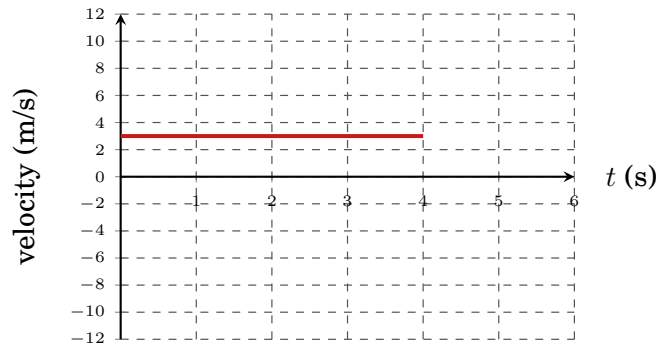
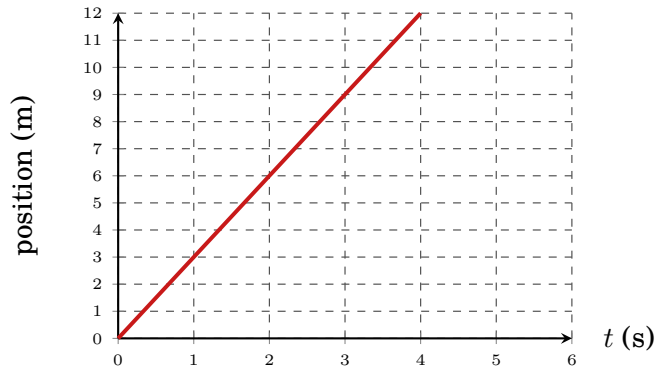
A and **B** are both straight lines sloping up. The only difference is **how steep** — and steepness on a position–time graph is speed. **C** slopes *down*, because the distance from the sensor is getting smaller; the walker has not gone anywhere negative, they have turned round.

the one to make them look at twice

D is a **horizontal line, and it is not on the axis**. Standing still means the position is not *changing*; it does not mean the position is zero. Expect a flat line drawn along the bottom, and it is worth catching: it is the same confusion as line 4 of the Do Now, in picture form.

1. Move with Constant Speed in + Direction

starts at 0 m • $v = +3 \text{ m/s}$ • $a = 0$



position-time

A **straight line sloping up**. The moped covers the same 3 m every second, so the position climbs by the same amount each second and the line never bends. The slope is +3 m per second — which is the velocity. It runs from 0 m to 12 m and takes 4 s to do it.

velocity-time

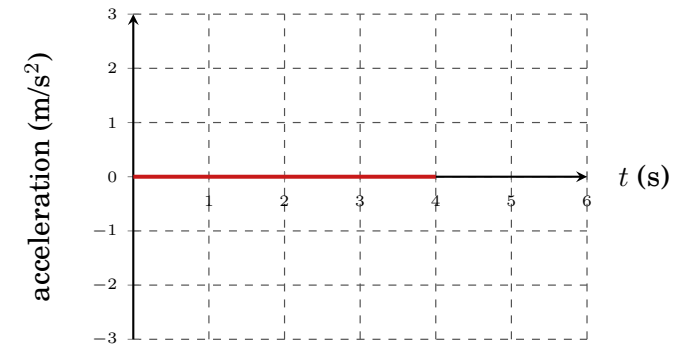
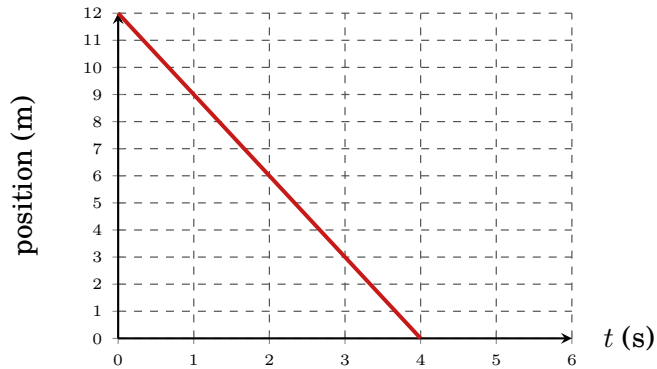
A **horizontal line above the axis**. The velocity is +3 m/s at the start, at the end, and at every instant in between, so the height never changes. It sits above the axis because the motion is in the + direction.

acceleration-time

A **horizontal line on the axis**. Constant velocity means the velocity is not changing, and acceleration *is* the rate at which velocity changes. Nothing is changing, so $a = 0$.

2. Move with Constant Speed in – Direction

starts at 12 m • $v = -3$ m/s • $a = 0$



position–time

A **straight line sloping down**. Same shape as motion 1, mirrored. It starts high because the moped starts at the far end of the track, and the position falls by 3 m every second. A downward slope on a p–t graph always means motion in the – direction.

velocity–time

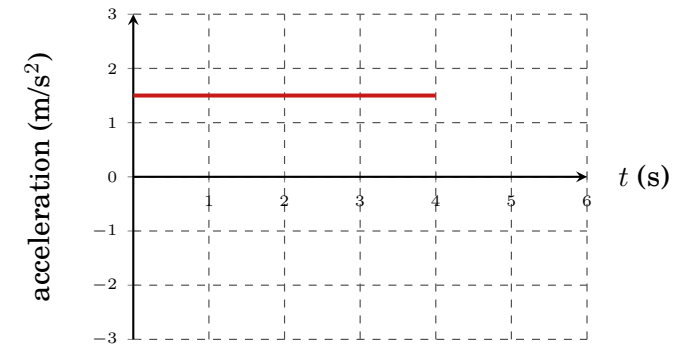
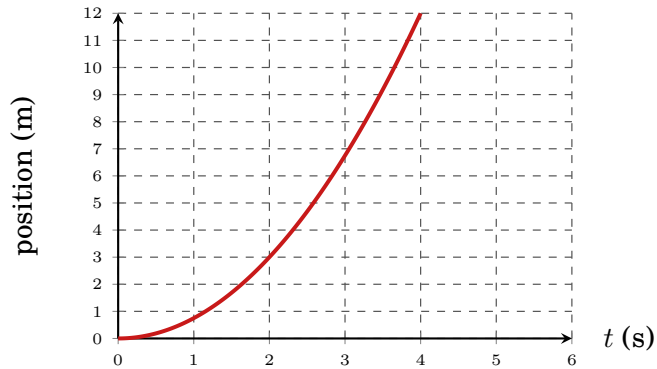
A **horizontal line below the axis**. Still constant, so still flat — but now at -3 m/s instead of $+3$. **The sign is the direction and nothing else.** The moped is going exactly as fast as in motion 1; it is simply going the other way.

acceleration–time

A **horizontal line on the axis**, exactly as in motion 1. Moving backwards is not accelerating. The velocity is unchanging, so the acceleration is zero.

3. Move in + Direction; Speeding Up

starts at 0 m • $v : 0 \rightarrow +6 \text{ m/s}$ • $a = +1.5 \text{ m/s}^2$



position-time

A curve that gets steeper. This is the first graph on the sheet that is not straight, and the bend is the whole point: a curve means the velocity is changing. It starts almost flat — the moped is barely moving — and steepens as the moped gains speed, because a steeper slope *is* a larger velocity.

velocity-time

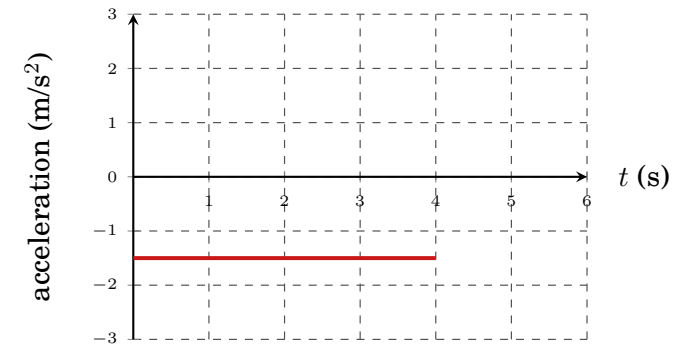
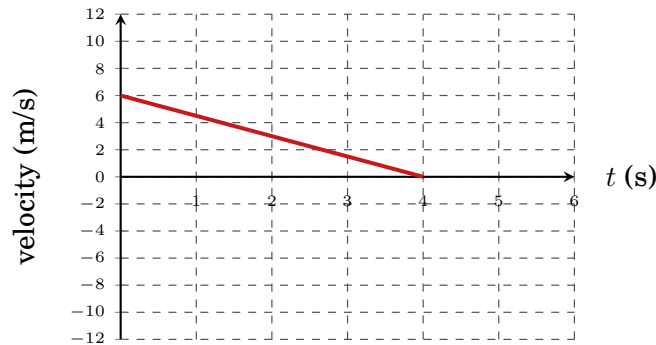
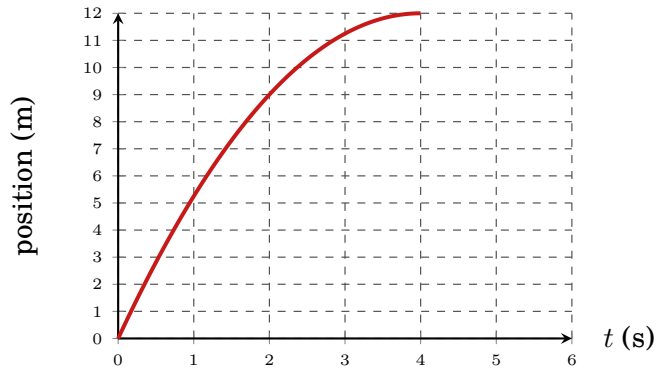
A straight line sloping up from the origin. The velocity starts at zero and grows by 1.5 m/s every second. Note that this graph is straight while the position graph is curved: they are describing the same motion, and they disagree in shape because they are plotting different quantities.

acceleration-time

A horizontal line above the axis, at +1.5. The velocity is changing, but it is changing at a *steady rate*, so the acceleration itself is constant. Above the axis because velocity and acceleration point the same way — and that is why it speeds up.

4. Move in + Direction; Slowing Down

starts at 0 m • $v : +6 \rightarrow 0 \text{ m/s}$ • $a = -1.5 \text{ m/s}^2$



position-time

A **curve that flattens out**. Still rising — the moped is still going forwards the whole time, so its position keeps increasing — but each second it covers less ground than the second before, so the line levels off. By $t = 4$ s it is horizontal: the moped has stopped.

velocity-time

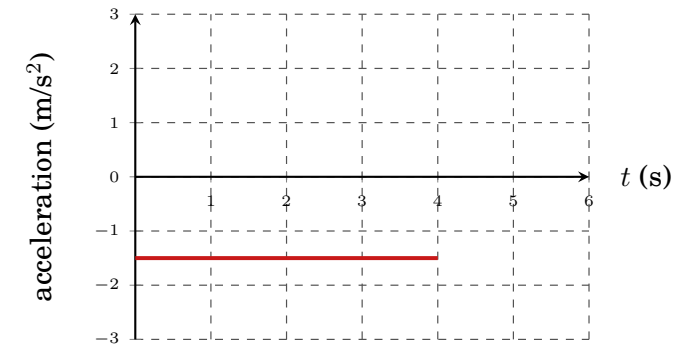
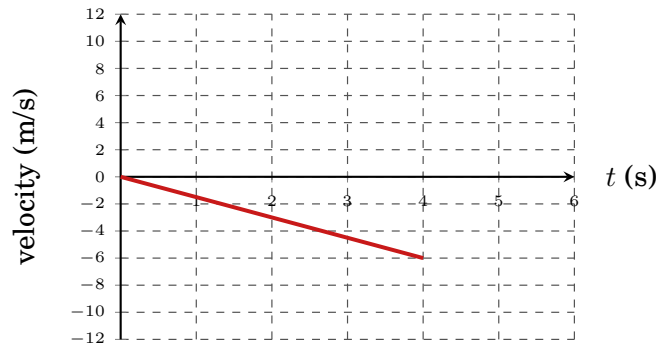
A **straight line sloping down to the axis**. It starts at $+6 \text{ m/s}$ and loses 1.5 m/s each second until it reaches zero. It stays *above* the axis throughout, because the moped never reverses — it only slows.

acceleration-time

A **horizontal line below the axis**, at -1.5 . Here is the trap: the moped is moving forwards, yet the acceleration is negative. Negative acceleration does not mean going backwards. It means the acceleration points backwards — against the motion — which is what slowing down is.

5. Move in – Direction; Speeding Up

starts at 12 m • $v : 0 \rightarrow -6 \text{ m/s}$ • $a = -1.5 \text{ m/s}^2$



position–time

A **curve that gets steeper, going down**. The mirror image of motion 3. It begins almost flat at the top of the graph and falls away ever more sharply as the moped picks up speed heading back towards zero.

velocity–time

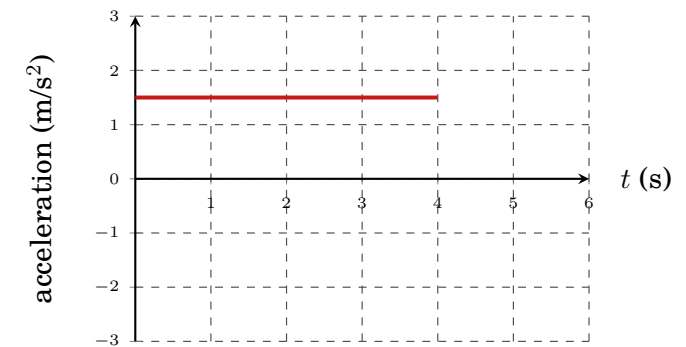
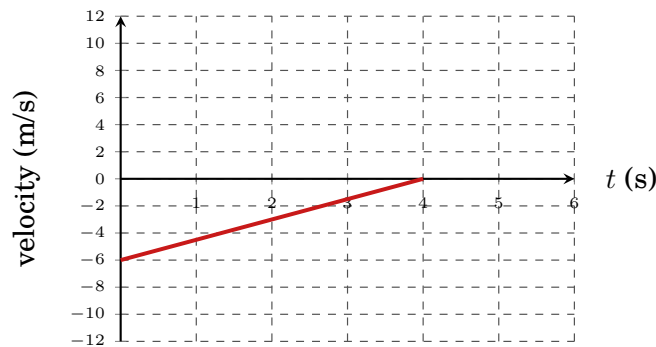
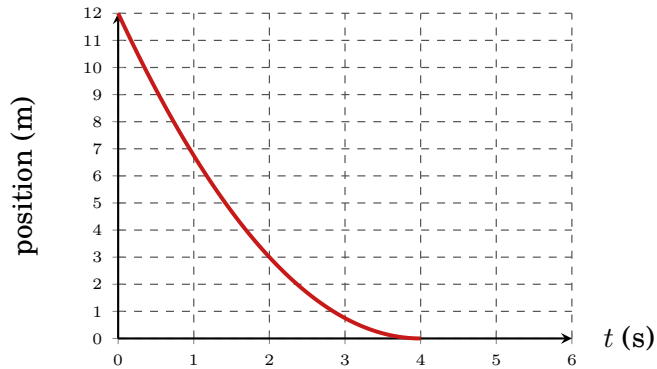
A **straight line sloping down from the origin, into the negative**. The velocity starts at zero and becomes more and more negative. **Read that carefully:** the *number* is getting smaller while the *speed* is getting larger, because speed is the size of the velocity and ignores the sign.

acceleration–time

A **horizontal line below the axis**, at -1.5 — the same as motion 4, which was *slowing down*. One acceleration graph, two opposite behaviours. The acceleration graph alone can never tell you whether something is speeding up. You have to look at the velocity graph with it.

6. Move in – Direction; Slowing Down

starts at 12 m • $v : -6 \rightarrow 0$ m/s • $a = +1.5$ m/s²



position-time

A **curve that flattens out, going down**. The moped is travelling backwards the whole time, so the position keeps falling — but by less and less each second, until the line goes horizontal and it stops at 0 m.

velocity-time

A **straight line rising to meet the axis from below**. It starts at -6 m/s and climbs to zero. The line goes *up* while the moped is *slowing down*, which looks wrong until you remember that height on this graph is velocity, not speed. It is getting less negative, and less negative means slower.

acceleration-time

A **horizontal line above the axis, at $+1.5$** . A positive acceleration on an object that never once moves forwards. Velocity is negative, acceleration is positive: opposite signs, so it slows down. This is the clearest case on the whole sheet that **the sign of the acceleration is not the direction of travel**.